



GE Medical Systems

Technical Publications

Direction 2153389

Revision 9

Signa® Release 8.X Block Diagrams & Supplemental Schematics

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785–5052/8*323–5052 **immediately** after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section “S” of the Policy & Procedure Bulletins.

3/12/92



GE Medical Systems

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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
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- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

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- 本维修手册仅存有英文本。
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- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
A	Mar 26, 1996	Preliminary release.
0	July 31, 1996	Expanded the Operator Workspace block diagrams, updated the PDU sections for Release 8.x changes, and made corrections and additions to other sections.
1	Nov 21, 1996	Corrections to E–Stop circuit, ISE Chassis temperature sensor circuit, and Compact PDU.
2	May 22, 1997	Incorporated information for Release 8.1 Video Conferencing and Cx Magnet Enclosure. Also made corrections/additions for CQAs/SPRs.
3	Sept. 3, 1997	Incorporated information for SGD Gradient Driver and RF/Pen II Cabinet.
4	Apr 23, 1998	Revised Operator Workspace Tab to include Octane Computer information. Added Multi–Nuclear Spectroscopy (1.5T) TAB and related information.
5	Dec 7, 1998	Incorporated 1.0T RF/PDU and Water Chiller information.
6	Feb 22, 1999	Added Power PC configuration to AP for 1.5T and 1.0T. Standardized Receiver naming convention to 0,1,2,3. (MRlge46464). Added information to include MR/i & CV/i circuitry to the Patient Handling Section, added pages 1–3 & 1–4. Added Sheet 4b UCERD information to SIGNA 8.X (1.0T & 1.5T) TPS/ISE/RF SUBSYSTEM. Updated Schematics in TPS/ISE (1.0T) & (1.5T) Sections.
7	Aug. 11, 1999	Changed System Section information, Patient Comfort Module to Front & Rear End Bell information and connections on page 1–4. Added information in Patient Handling Section, to include MR/i & CV/i circuitry, added page 1–3 which changed the existing page 1–3 to 1–4. Removed the word HORIZON from all headers on all pages. Added new Bit 3 hardware, in Sections ; SYSTEM, OPERATOR WORKSPACE, TPS/ISE 1.5T, and TPS/ISE 1.0T.
8	Oct 19, 2000	Changed MNS Section information. Updated the Overview tab Section 3 Abbreviations. Changed microphone and added ACGD and SRF graphics. Removed the word 'Horizon' from any pages it appeared on. Added ACGD Cabinet information Page 3–1 to the Gradient Tab.
9	Nov 22, 2005	TPS/ISE Tabs (1.5T & 1.0T) – Corrected typo J34 to J41 on the REAR I/F BOARD Page 1–8. Added Note3 to page 1–14 for referring to surface coil service documentation for troubleshooting information (MRlge93458).

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* This revision number/letter corresponds to the indicated document's revision control system.

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OVERVIEW

Purpose
Standard Conventions
Abbreviations



SYSTEM

Overall System Block Diagrams



OPERATOR WORKSPACE

Operator Workspace Block Diagrams



GRADIENT

Gradient Block Diagrams (SGD)
Gradient Block Diagrams (8645)
Gradient Block Diagram (ACGD)



PATIENT HANDLING

Patient Handling Block Diagrams



PDU

PDU Block Diagrams



COMPACT PDU

Compact PDU Block Diagrams



TPS/ISE (1.5T)

TPS/ISE Block Diagrams (1.5T)
RF Supplemental Schematics (1.5T)



TPS/ISE (1.0T)

TPS/ISE Block Diagrams (1.0T)
RF Supplemental Schematics (1.0T)



MULTI-NUCLEAR SPECTROSCOPY (1.5T)

MNS Block Diagrams and Interconnects (1.5T)
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